Tucker Johnson

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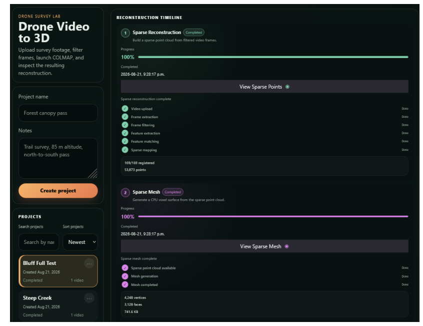
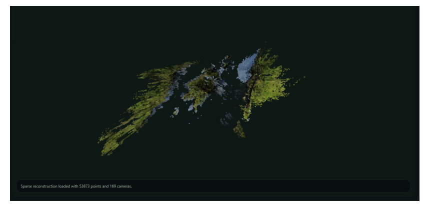
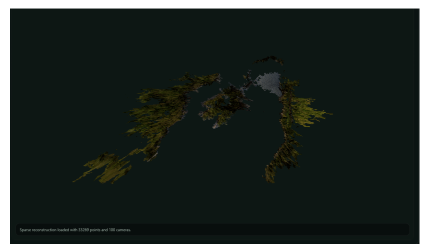
Portfolio

Drone Video to 3D Simulation Software



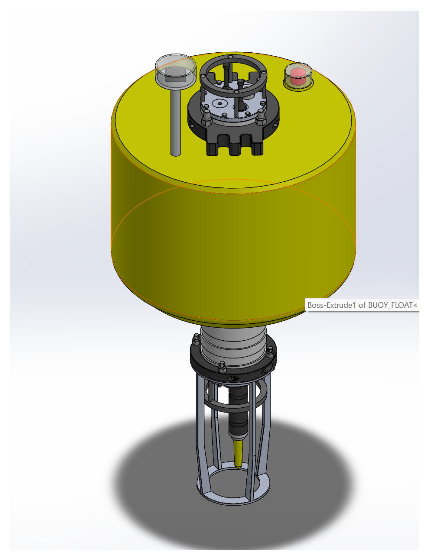


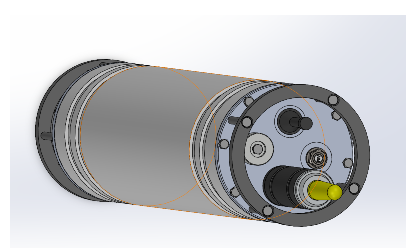


- Developed a full-stack photogrammetry platform that transforms raw drone video into interactive 3D reconstructions, providing a complete workflow from video upload and processing through visualization, analysis, and export.
- Built an automated computer-vision pipeline using FFmpeg, OpenCV, and COLMAP for frame extraction, image-quality filtering, feature detection and matching, camera pose estimation, sparse reconstruction, and CUDA-accelerated dense reconstruction.
- Implemented multiple reconstruction outputs including sparse and dense point clouds, CPU-generated sparse meshes, and Poisson dense meshes, with dedicated processing timelines and quality controls for managing each stage.
- Developed an interactive React and Three.js 3D environment for exploring reconstructed scenes, including camera trajectories, real-world scaling, distance and area measurements, elevation analysis, point inspection, annotations, and point-cloud cleanup tools.
- Designed a Smart Retry system that analyzes image registration, feature coverage, verified-match connectivity, reconstruction quality, and previous retry results to diagnose failed or partial reconstructions and selectively rerun only necessary processing stages.
- Built a FastAPI and SQLite backend to manage projects, reconstruction jobs, processing state, artifacts, and reconstruction history, with tools for comparing runs and exporting point clouds, meshes, measurements, maps, and reconstruction reports.

Technologies: Python · TypeScript · React · Three.js · FastAPI · SQLite · OpenCV · FFmpeg · COLMAP · CUDA

Marine Monitoring Buoy

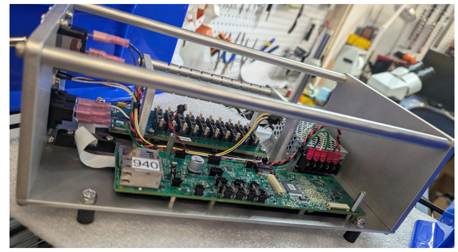




- Designed a conceptual autonomous marine monitoring buoy in SolidWorks for localized fish and lobster habitat monitoring using underwater acoustic sensing.
- Developed a sealed underwater pressure housing with O-ring endcaps, providing protected internal space for the battery system, electronics chassis, and data-storage hardware.
- Designed a modular internal layout incorporating a high-capacity battery pack, WiFi PCB, power and control electronics, and multiple SD cards for onboard acoustic data storage.
- Integrated the hydrophone, pressure relief valve (PRV), electrical connections, and anode into the endcap design while considering sealing, component placement, and accessibility.
- Designed the surface buoy and supporting structure to maintain the underwater housing while incorporating a navigation light, WiFi communication hardware, and protection for the hydrophone.
- Created the complete buoy as a multi-component SolidWorks assembly, using individual part models and subsystem integration to evaluate packaging, fit, component clearances, and the overall mechanical architecture.

SD Card & Battery Pack Test Fixtures

- Designed and developed SD card and battery pack test fixtures to standardize electrical testing and provide repeatable, easy-to-use test setups.
- Integrated existing electrical components including PCBs, switches, power and Ethernet connectors, SD card hardware, banana jacks, LEDs, and voltage displays into custom fixture designs.
- Designed custom mounting plates, supports, enclosures, and complete assemblies in SolidWorks, producing detailed part and assembly drawings for manufacturing.
- Created electrical wiring diagrams for both fixtures, defining component connections and supporting assembly, testing, and future troubleshooting.
- Manufactured and assembled the fixtures through drilling, soldering, wiring, component mounting, and mechanical assembly, while sourcing and ordering required components.
- Validated both completed fixtures through electrical continuity and connection checks, functional testing, and verification of SD card and battery pack testing operations.



NavBot - Autonomous Medication Delivery System

- Developed NavBot, an autonomous medical delivery robot designed to reduce repetitive hospital tasks by transporting medication along trained routes while providing autonomous navigation and obstacle-safety features.
- Designed and constructed a rigid VEX-based chassis and drivetrain, iterating the frame, wheel supports, and component mounting to improve stability, weight distribution, and driving performance.
- Designed a motor-actuated medication box system with a hinged lid, lever-style opening mechanism, and secondary motorized locking mechanism to securely store medication and automatically provide access at delivery locations.
- Programmed Training and Autonomous modes in C++, allowing operators to manually train routes while recording drive distances and headings, then autonomously replay stored routes for medication delivery.
- Implemented Dijkstra's shortest-path algorithm with an adjacency-matrix representation of stored routes, allowing the robot to determine and execute the shortest available route between hospital locations.
- Implemented closed-loop heading control using proportional and PI control, dynamically correcting motor output during straight-line driving and turns to reduce drift and improve autonomous path accuracy.
- Integrated distance, bumper, touch, inertial, and motor feedback into the control system for obstacle detection, collision response, heading control, and automated medication-box operation.

Technologies: C++ · VEX IQ · SolidWorks · Dijkstra's Algorithm · PI/P Control · Sensor Integration · Embedded Robotics